Stochastic economies and estimation

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Introduction

- **Workshop so far**: MIT shocks (nonlinear and linear)
- Fine object for very rare events, but for business-cycle-like shocks, **rational expectations** is more useful benchmark
  - How useful are MIT shocks for rational expectations solution?
- Our answer (over today and tomorrow's lectures): **very useful!**
  - In fact, for practical purposes, **MIT shocks are all you need!** ("certainty correspondence")
  - Today: first instance of this: MIT shocks are all you need **to first order** ("certainty equivalence"). Apply this to **simulations** + **estimation**.

First-order certainty equivalence

Setup

- Consider general macro model:

$$F\left(X_{t-1}, X_t, \mathbb{E}_t\left[X_{t+1}\right], \epsilon_t\right) = 0$$

  - $X_t \in \mathbb{R}^n$ are endogenous variables [het-agent applications: n large, eg $n = 2{,}000$]
  - $\epsilon_t \in \mathbb{R}$ are exogenous innovations (realized shock process) [naturally extends to >1 shocks]
    - Distributed $\epsilon_t = \sigma_R \cdot \bar{\epsilon}_t$ with $\bar{\epsilon}_t$ iid, mean 0, variance 1, and symmetric density
- Expectations $\mathbb{E}_t[\,\cdot\,]$ are taken assuming future distribution of ϵ_t is $\sigma \cdot \bar{\epsilon}_t$
  - **Rational expectations**: $\sigma = \sigma_R$ (perceived distribution of future ϵ_t = truth)
  - **Perfect foresight**: $\sigma = 0$ (agents think all future ϵ_t will realize to 0 for sure)
- **Deterministic steady state** $X \in \mathbb{R}^n$ solves $F(X, X, X, 0) = 0$ (i.e. $\sigma = \sigma_R = 0$)

PE SIM example

- For concreteness, consider the PE SIM model

$$V_t(e,a) = \max_{c,a'} u(c) + \beta \mathbb{E}_{e'}[V_{t+1}(e',a') \,|\, e]$$

$$\text{s.t.} \qquad a' + c = (1+r)a + Z_t e \qquad a' \geq 0$$

- Aggregate income Z_t follows AR(1) process: $Z_t - 1 = \rho(Z_{t-1} - 1) + \epsilon_t$
- Here, $X_t \equiv (\boldsymbol{v}_t, \boldsymbol{\phi}_t, C_t, Z_t)$ where $\boldsymbol{v}_t$ stores marginal value $V_{a,t}$ over gridpoints (e,a), $\boldsymbol{\phi}_t$ stores the p.d.f, and C_t aggregates consumption policy over distribution
- Equations describing F: the law of motion of Z_t, plus

Dynamic programming: expectations enter via $\mathbb{E}_t[v_{t+1}]$

$$\boldsymbol{v}_t = \mathscr{V}(Z_t, \mathbb{E}_t \boldsymbol{v}_{t+1}) \qquad \boldsymbol{\phi}_t = \Lambda(Z_t, \mathbb{E}_t \boldsymbol{v}_{t+1}, \boldsymbol{\phi}_{t-1}) \qquad C_t = \mathscr{C}(Z_t, \mathbb{E}_t \boldsymbol{v}_{t+1}, \boldsymbol{\phi}_{t-1})$$

(Stochastic) sequence space solution

- Given σ, consider **nonlinear sequence-space solution** to F:

$$X_t = \mathscr{X}\left(\sigma; \epsilon_t, \epsilon_{t-1}, \epsilon_{t-2}, \ldots\right)$$ [Lan-Meyer-Gohde, Lombardo-Uhlig]

[eg, obtained from stable state-space solution $X_t = g\left(X_{t-1}, \epsilon_t; \sigma\right)$]

- We are going to relate this to the **perfect-foresight solution**

$$X_t^{PF} = \mathscr{X}\left(0; \epsilon_t, \epsilon_{t-1}, \epsilon_{t-2}, \ldots\right)$$

[eg, obtained from ∞-long sequence of repeated-surprise MIT shocks]

- Use **perturbation approach**: linearize $\mathscr{X}$ around $\sigma = \sigma_R = 0$ (i.e. $\boldsymbol{\epsilon} = \mathbf{0}$)
- Helpful result: $\mathscr{X}\left(\sigma; \epsilon_t, \epsilon_{t-1}, \ldots\right) = \mathscr{X}\left(-\sigma; \epsilon_t, \epsilon_{t-1}, \ldots\right)$, so, e.g., $\mathscr{X}_\sigma(\mathbf{0}) = 0$

MIT shock impulse response

- Special case of X^{PF} is the MIT shock impulse response

$$X_j^{MIT}(\nu) \equiv \mathscr{X}(0; 0, \ldots, 0, \underset{\substack{\uparrow \\ \text{position } j}}{\nu}, 0 \ldots)$$

- Can compute $\{X_j^{MIT}(\nu)\}_j$ with nonlinear perfect foresight starting from s.s.
  - cf lecture 1 and 3 methods!
- **Next**: why this is sufficient to get the first-order solution with aggregate risk!

Certainty equivalence

- Expand $X_t = \mathscr{X}\left(\sigma, \epsilon_t, \epsilon_{t-1}, \ldots\right)$ to first order in $(\sigma, \boldsymbol{\epsilon})$:

No anticipated risk,
no past realization

$$X_t = X + \cancel{\mathscr{X}_\sigma}\sigma + \frac{\partial \mathscr{X}}{\partial \epsilon_0}(\mathbf{0})\,\epsilon_t + \frac{\partial \mathscr{X}}{\partial \epsilon_{-1}}(\mathbf{0})\,\epsilon_{t-1} + \cdots + \mathcal{O}(\sigma^2)$$

Symmetry:
Steady state unchanged

- So, to first order, X_t follows **linear MA** process, with coefficients equal to

$$\frac{\partial \mathscr{X}}{\partial \epsilon_{-j}}(\mathbf{0}) = \frac{dX_j^{MIT}}{d\nu}(0) \equiv \mathscr{X}_j$$

Certainty equivalence. To first order, X_t follows a linear MA process whose coefficients are given by the first order MIT shock impulse response of X to ϵ [Boppart-Krusell-Mitman]

Implementation using SSJ

- To first-order, X_t (so all its elements!) follows linear $MA(\infty)$

$$X_t = X + \mathscr{X}_0 \epsilon_t + \mathscr{X}_1 \epsilon_{t-1} + \mathscr{X}_2 \epsilon_{t-1} + \cdots + \mathcal{O}(\sigma^2) \qquad \mathscr{X}_j \equiv \frac{dX_j^{MIT}}{d\nu}(0)$$

- Convenient: we can use SSJ to very effectively get $\mathscr{X}_j$!

- For instance, suppose that we want the behavior of aggregate outcome X_t to monetary shock r_s in the canonical HANK model, with $MA(\infty)$ shock process

$$r_t = r + \mathscr{R}_0 \epsilon_t + \mathscr{R}_1 \epsilon_{t-1} + \mathscr{R}_2 \epsilon_{t-1} + \cdots +$$

- Then we get the SSJ $G^{X,r}$ of X to r, then apply vector-matrix multiplication

$$(\mathscr{X}_0, \mathscr{X}_1, \mathscr{X}_2, \cdots)' = G^{X,r} \cdot (\mathscr{R}_0, \mathscr{R}_1, \mathscr{R}_2, \cdots)'$$

Applications: simulation, second moments, estimation

Application to simulation

- Suppose we want to do stochastic simulations of X_t. Since

$$X_t \simeq X + \mathscr{X}_0 \epsilon_t + \mathscr{X}_1 \epsilon_{t-1} + \mathscr{X}_2 \epsilon_{t-1} + \cdots \qquad (\star)$$

- Once we have $\mathscr{X}_t$, we can just simulate by applying $(\star)$!

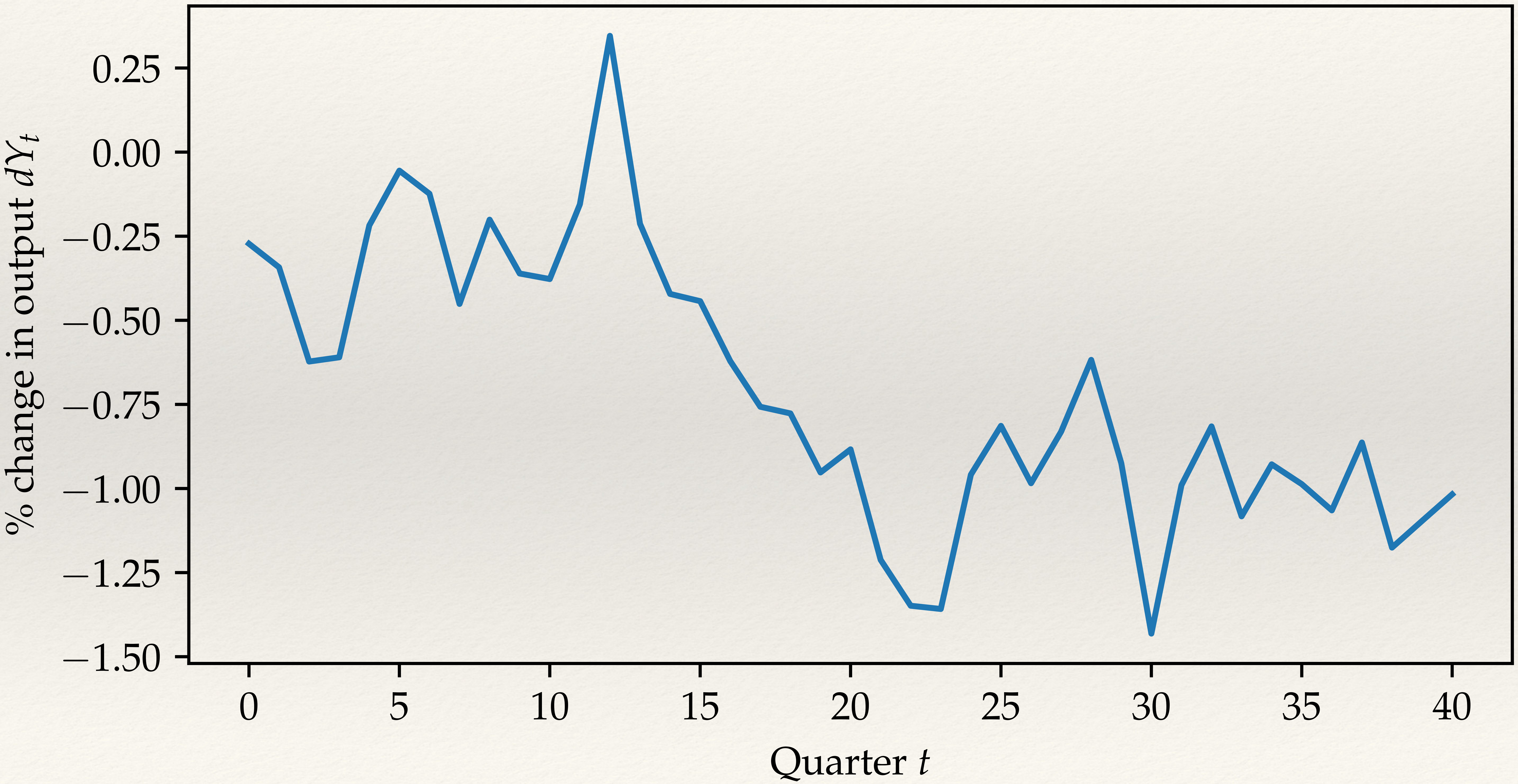
```python
rho = 0.9
dishock = rho**np.arange(T)
dY = G['Y', 'ishock'] @ dishock
```

✓ 0.0s

```python
sigma = 0.005
Tsim = 1_000_000                    # will simulate for 1 million quarters
np.random.seed(40)
eps = np.random.randn(Tsim+T)   # draw epsilons for T additional pre-per
dY_sim = np.empty(Tsim)
for t in range(Tsim):
    # at each t, take window of current up to T-1 previous epsilons
    # take dot product with MA coefficients implied by MIT shock
    dY_sim[t] = np.dot(dY, eps[t:t+T][::-1]) * sigma
```

✓ 0.7s Python

Simulating output response to monetary shock



Application to second moments

- Suppose we want to obtain second moments of X_t (with $\sigma_R = \sigma$). Have

$$X_t \simeq X + \mathcal{X}_0\epsilon_t + \mathcal{X}_1\epsilon_{t-1} + \mathcal{X}_2\epsilon_{t-1} + \cdots \qquad (\star)$$

and we can just apply time series formulas to get these!

- For instance ergodic variance is area under squared impulse response

$$\mathrm{Var}(X_t) = \sigma^2 \sum_{j=0}^{\infty} \left(\mathcal{X}_j\right)^2$$

- More generally

$$\mathrm{Cov}(X_t, X_{t-k}) = \sigma^2 \sum_{j=0}^{\infty} \mathcal{X}_j \mathcal{X}_{j+k}$$

Implementation: second moments in HANK model

```
var_dY = sigma**2 * (dY @ dY)
```

✓ 0.0s Python

Compare to the sample variance to make sure this is right. Note that it's close but not exact, reflecting Monte Carlo error:

```
r_dY = (dY_sim - dY_sim.mean()) @ (dY_sim - dY_sim.mean()) / Tsim
ample variance: {sample_var_dY:.4e} | theoretical : {var_dY:.4e}")
```

✓ 0.0s Python

```
Sample variance: 3.1761e-05 | theoretical : 3.1870e-05
```

Application to impulse-response matching

- Straightforward to estimate by **matching impulse responses**
- Suppose model parameters are θ, giving impulse response $IRF_t(\theta)$, and identified empirical impulse response $\hat{irf}_t$
- Then we can just search for

$$\hat{\theta} = \arg\min \left(IRF(\theta) - \hat{irf} \right)' V^{-1} \left(IRF(\theta) - \hat{irf} \right)'$$

where, e.g. V has sample variances of $\hat{irf}_t$ (Christiano, Eichenbaum, Evans)

- This is especially fast if we do not have to recompute all Jacobians for $IRF(\theta)$, such as we estimate shock process parameters (G unchanged!)

Application to method of moments

- Also straightforward to estimate by **matching moments**
- Suppose model parameters are θ, giving model moments $M(\theta)$
  - eg second moments, HP-filtered moments, regression coefficients…
- Then we can confront to empirical counterparts $\hat{m}$ by taking

$$\hat{\theta} = \arg\min \left(M(\theta) - \hat{m}\right)' V^{-1} \left(M(\theta) - \hat{m}\right)'$$

where, e.g. V is model's $\mathbb{E}\left[M(\theta)\, M(\theta)'\right]$

(Minimum distance estimation, aka "simulated" method of moments)

Application to likelihood-based estimation

- Let's assume that model innovations ϵ_t are normally distributed
- Given parameter θ, first-order process followed by X_t is

$$X_t \simeq X(\theta) + \mathcal{X}_0(\theta)\epsilon_t + \mathcal{X}_1(\theta)\epsilon_{t-1} + \mathcal{X}_2(\theta)\epsilon_{t-1} + \cdots$$

- Given data $\mathbf{X}$, we know the **(log) likelihood function**

$$\mathcal{L}(\mathbf{X};\theta) = -\frac{1}{2}\log\det\mathbf{V}(\theta) - \frac{1}{2}\mathbf{X}'\mathbf{V}(\theta)^{-1}\mathbf{X}$$

where $\mathbf{V}(\theta)$ is variance-covariance matrix of X_t at all lags

- In practice, use Choleski decomposition to calculate $\det\mathbf{V}$ and $\mathbf{X}'\mathbf{V}^{-1}\mathbf{X}$

Application to likelihood-based estimation

- Given likelihood function $\mathscr{L}(\mathbf{X};\theta)$ we can in turn do:
  - Maximum likelihood estimation

$$\hat{\theta} = \arg\max \mathscr{L}\left(\boldsymbol{Y}, \theta\right)$$

  - Bayesian estimation given prior $p\left(\theta\right)$: find posterior distribution

$$p\left(\theta|\,\mathbf{Y}\right) = \frac{\mathscr{L}\left(\mathbf{Y}\,|\,\theta\right)p\left(\theta\right)}{p\left(\mathbf{Y}\right)} \propto \mathscr{L}\left(\mathbf{Y}\,|\,\theta\right)p\left(\theta\right)$$

- For instance, posterior mode just solves

$$\hat{\theta} = \arg\max p\left(\theta\,|\,\mathbf{Y}\right)$$

[More complex: simulate from posterior using MCMC]

Practical implementation: estimate HANK model!

- Notebook estimates HANK model with
  - 3 shocks (government spending, TFP, monetary policy)
  - 3 observables (output, inflation, interest rates)
- Look for
  - shock process parameters (assumed AR(1)'s: 3 σ + 3 ρ)
  - coefficient on Taylor rule ϕ_π, fiscal rule ϕ_τ, slope of Phillips curve κ_w
- Done in less than a minute on a laptop!

Conclusion

- MIT shocks from SSJ are all you need for first-order solution!
  - Simulation, model moments, estimation
- Obtain same answer as state-space models linearized to first order
  - But speed gains can be significant
- More to come in next lecture, and then tomorrow: beyond certainty equivalence, MIT shocks are still all you need!